Assignment #2: Design a Three-Page Website Using HTML and CSS

Course: Web Technologies — BSCS, Semester 5
Total Marks: 100
Mode: Individual Assignment

Objective

Develop a website with **three distinct, linked HTML pages**, demonstrating semantic HTML, multimedia, tables, forms, and three methods of applying CSS.

1. Choose One Website Topic

Topic	Page 1: Home	Page 2: Information	Page 3: Form
Personal Portfolio	Introduction and skills	Projects and services	Project inquiry
Training Academy	Introduction and courses	Course schedule and fees	Student registration
University Society	Introduction and activities	Events and membership plans	Membership registration
Travel Agency	Introduction and destinations	Tour packages and prices	Booking request

Use sample information appropriate to your chosen topic.

2. Three Pages and Their CSS Methods

File	Required CSS method	Instructions
`index.html`	**Inline CSS**	Apply styles using the `style` attribute on HTML elements.
`information.html`	**Embedded CSS**	Write CSS rules inside a `<style>` element in the document's `<head>`.
`registration.html`	**External CSS**	Link to a separate file named `style.css`.

Use **only the assigned CSS method on each page**. Maintain a consistent color scheme and readable typography across all three pages.

3. Page Requirements

Page 1 — Home: Inline CSS

Include:

- `<header>` containing the website title.
- `<nav>` with working links to all three pages.
- `<main>` with at least two `<section>` elements, each with a heading.
- A relevant image with descriptive `alt` text.
- A list of at least four skills, courses, activities, or destinations.
- A `<footer>` with suitable contact or copyright information.

Use inline CSS to style text, backgrounds, spacing, and the image.

Page 2 — Information: Embedded CSS

Include:

- Navigation links to all three pages.
- An `<article>` containing a self-contained description of a project, course, event, or tour.
- An `<aside>` containing related information, such as an announcement or useful tip.
- A table with a descriptive caption, **at least three columns and four data rows**, and appropriate header cells.
- One audio clip and one video clip, both with playback controls and fallback text.

Use embedded CSS to style headings, the article, aside, and table. Give the table visible borders, cell padding, and a distinct header background.

Page 3 — Form: External CSS

Create a registration, inquiry, or booking form appropriate to your topic. Include an account-creation area to demonstrate the password field.

The form must contain:

- Full Name: text input.
- Email: email input.
- Password: password input.
- Age: number input with a suitable minimum and maximum.
- Preferred Date: date input.
- A radio-button group with at least two options and only one possible selection.
- At least three independent checkboxes.
- A dropdown with a placeholder and at least three selectable options.
- A `<textarea>` for additional information.
- Submit and Reset buttons.

Use `<fieldset>` and `<legend>` to group related controls. Give controls suitable labels, unique IDs, and meaningful `name` attributes.

Apply these validation attributes appropriately:

- `required` for Full Name, Email, Password, and the dropdown.
- `minlength="8"` for Password.
- `min` and `max` for Age.
- `maxlength` for the comments field.

Use external CSS to style labels, inputs, fieldsets, buttons, and spacing. Demonstrate `:focus` on inputs and `:hover` on buttons or links.

4. CSS Selector Requirements

Across the **embedded and external stylesheets**, demonstrate all four selector types:

Selector	Example
Element selector	`p { color: navy; }`
Class selector	`.notice { background-color: lightyellow; }`
ID selector	`#registration-form { padding: 20px; }`
Attribute selector	`input[type="email"] { border: 1px solid gray; }`

Each selector must match an element and produce a visible effect. Add a CSS comment identifying each selector type.

Use suitable CSS properties for colors, fonts, backgrounds, borders, margins, padding, and dimensions.

5. General Requirements and Testing

- Use a complete HTML5 document structure and a meaningful title on every page.
- Use semantic elements for meaningful content. Use `<div>` or `<span>` only where a generic grouping or styling container is appropriate.
- Add useful HTML comments explaining major parts of the code.
- Ensure navigation links, images, audio, and video work using relative file paths.
- Validate all three HTML pages using the **W3C Markup Validation Service** at https://validator.w3.org/. Correct errors and review warnings.
- Test form validation using empty required fields, an invalid email, and a password shorter than eight characters.
- Test the website in two browsers and record any differences.

This is a front-end assignment. A database, login system, and server-side form processing are not required. Use dummy details when testing the form.

6. Short Explanation

Include a **200–300-word explanation** in `README.md` covering:

- What CSS is and its benefits.

- The selector, property, and value in one rule from your website.
- The differences between inline, embedded, and external CSS.
- A brief overview of CSS1, CSS2/2.1, and modern modular CSS.
- Why browser support matters, including the two browsers you tested.
- One example of semantic HTML and one example of non-semantic HTML from your code.

7. Submission

Push the complete project to GitHub and publish it through GitHub Pages. Include:

- Three HTML files and `style.css`.
- All required media files.
- `README.md`.
- Validation screenshots for all three HTML pages.

Enter your **name, roll number, section, GitHub repository URL, and live website URL** in the shared Google Sheet.

Marking Rubric

Criterion	Assessment details	Marks
Website structure and navigation	Three distinct pages (3); working navigation across all pages (3); complete document structures and meaningful titles (2)	8
Semantic HTML and comments	Appropriate use of all required semantic elements (7); sensible generic containers (1); useful HTML comments (2)	10
Images and lists	Relevant image with descriptive alternative text (3); correctly structured list (2)	5
Table	Required rows and columns (2); caption and header cells (2); correct structure (1)	5
Audio and video	Working audio with controls and fallback text (3); working video with controls and fallback text (3)	6
Form controls	Text, email, password, number, and date inputs (5); radio group (2); checkboxes (2); dropdown (1); textarea (1); submit/reset buttons (1); labels, names, IDs, and grouping (3)	15
Form validation	Required fields (2); password minimum length (1); age limits (1); comments maximum length (1)	5
Three CSS methods	Correct inline styling on Page 1 (5); embedded styling on Page 2 (5); external stylesheet linked and applied on Page 3 (5)	15
CSS selectors	Working element, class, ID, and attribute selectors, identified with comments (2 marks each)	8
Visual presentation	Consistent colors and readable text (3); spacing and borders (3); table and form styling (3); hover and focus feedback (2)	11
Validation and	W3C validation evidence for three pages with errors corrected (3);	4

browser testing	documented testing in two browsers (1)	
Written explanation	Accurate coverage of the specified CSS and HTML concepts	5
Submission	Complete repository (1); working live website (1); correct Google Sheet entry (1)	3
Total		**100**

Partial credit will be awarded for partially completed requirements. Be prepared to explain your HTML structure and CSS rules.